Gradient Descent and Optimization

Section 1: Loss Function Derivatives and Stationary Points

1 Find the stationary point (global minimum) of the scalar mean squared error loss $L(w) = 3(w-4)^2 + 7$.

a. $w = 4$

b. $w = 7$

c. $w = 3$

d. $w = 0$

2 For the loss function $L(w) = w^4 - 4w^2 + 6$, find all values of w where the gradient is zero.

a. $w = 0, \pm\sqrt{2}$

b. $w = \pm 2$

c. $w = 0, \pm 2$

d. $w = \pm 1$

3 In optimization, a critical point has $L'(w) = 0$ and $L''(w) = -8$. What type of point is this?

a. Local Maximum

b. Local Minimum

c. Saddle Point

d. Inflection Point

Section 2: Multivariate Gradients and Directional Derivatives

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4 Compute the gradient vector ∇L of $L(w_1, w_2) = 3w_1^2 + 5w_2^2 - 2w_1w_2$ evaluated at $(w_1, w_2) = (2, 1)$.

a. $[10, 6]^T$

b. $[12, 10]^T$

c. $[8, 8]^T$

d. $[14, 4]^T$

5 In which direction does the negative gradient vector $-\nabla L(\mathbf{w})$ point?

a. The direction of steepest decrease of the loss function

b. The direction of steepest increase of the loss function

c. Orthogonal to the level curves in the direction of highest loss

d. Along the tangent line of the contour

6 Compute the directional derivative of $f(x, y) = x^2y$ at $(1, 2)$ in the unit direction $\mathbf{u} = [0.8, 0.6]^T$.

a. 3.8

b. 4.2

c. 2.6

d. 5.0

Section 3: Gradient Descent Step Updates

7 A model weight has value $w_0 = 5.0$. If $\frac{\partial L}{\partial w} = 4.0$ and learning rate is $\eta = 0.1$, find the updated weight w_1.

a. 4.6

b. 5.4

c. 4.0

d. 4.9

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8 Consider the quadratic loss $L(w) = \frac{1}{2}aw^2$ with $a = 10$. What is the maximum learning rate η before gradient descent diverges?

a. $\eta < 0.2$

b. $\eta < 0.1$

c. $\eta < 0.05$

d. $\eta < 1.0$

9 Perform one vector gradient descent update from $\mathbf{w}_0 = [1, 2]^T$ with $\nabla L = [2, -4]^T$ and $\eta = 0.25$.

a. $[0.5, 3.0]^T$

b. $[1.5, 1.0]^T$

c. $[0.5, 1.0]^T$

d. $[1.0, 3.0]^T$

Section 4: Momentum and Advanced Optimizers

10 In Polyak momentum, the velocity update is $\mathbf{v}_{t+1} = \beta \mathbf{v}_t + \eta \nabla L$. If $\beta = 0.9$, $\mathbf{v}_0 = 2.0$, $\eta = 0.1$, and $\nabla L = 4.0$, find $\mathbf{v}_1$.

a. 2.2

b. 2.0

c. 1.8

d. 2.6

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11 Why does the Adam optimizer maintain running exponential averages of both first moments (mean) and second raw moments (uncentered variance)?

a. To adapt per-parameter learning rates inversely proportional to gradient magnitude

b. To guarantee convergence to the global minimum in non-convex losses

c. To replace backpropagation with finite differences

d. To eliminate the need for mini-batch sampling

12 What is the condition on the Hessian matrix $\mathbf{H} = \nabla^2 L(\mathbf{w})$ for a loss function to be strictly convex everywhere?

a. $\mathbf{H}$ is positive definite everywhere (all eigenvalues > 0)

b. $\mathbf{H}$ has determinant equal to zero

c. $\mathbf{H}$ has trace equal to 1

d. $\mathbf{H}$ is skew-symmetric

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